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Studierichting: Informatica
Oefeningenreeks 4A nr. 46.2

$$\int \frac{-3x+3}{2x^4+5x^2+2}dx \overset{u=x^2}{\Rightarrow} u = \frac{-5 \pm \sqrt{5^2-4(2)(2)}}{2(2)} = \left\{-2, -\frac{1}{2}\right\} \Rightarrow \frac{1}{2}\int \frac{-3x+3}{(x^2+2)\left(x^2+\frac{1}{2}\right)}dx =$$

$$\frac{1}{2}\int \frac{Ax+B}{(x^2+2)}dx + \frac{1}{2}\int \frac{Cx+D}{\left(x^2+\frac{1}{2}\right)}dx$$

$$\Rightarrow \begin{array}{lll} A\left(i\sqrt{2}\right)+B = & \left.\dfrac{-3x+3}{\left(x+\dfrac{1}{2}\right)}\right|_{x=i\sqrt{2}} & = -2+i(2\sqrt{2}) \Rightarrow A=2, B=-2 \\ C\left(\dfrac{i}{\sqrt{2}}\right)+D = & \left.\dfrac{-3x+3}{(x^2+2)}\right|_{x=\frac{i}{\sqrt{2}}} & = 2-i\sqrt{2} \Rightarrow C=-2, D=2 \end{array}$$

$$\Rightarrow \frac{1}{2}\int \frac{2x-2}{x^2+2}dx + \frac{1}{2}\int \frac{-2x+2}{x^2+\frac{1}{2}}dx =$$

$$\frac{1}{2}\int \frac{d(x^2+2)}{x^2+2} - \frac{1}{\sqrt{2}}\int \frac{d\left(\frac{x}{\sqrt{2}}\right)}{\left(\frac{x}{\sqrt{2}}\right)^2+1} - \frac{1}{2}\int \frac{d\left(x^2+\frac{1}{2}\right)}{x^2+\frac{1}{2}} + \sqrt{2}\int \frac{d\left(x\sqrt{2}\right)}{(x\sqrt{2})^2+1}$$

$$= \frac{1}{2}\ln|x^2+2| - \frac{1}{\sqrt{2}}\operatorname{Bgtan}\left(\frac{x}{\sqrt{2}}\right) - \frac{1}{2}\ln\left|x^2+\frac{1}{2}\right| + \sqrt{2}\operatorname{Bgtan}\left(x\sqrt{2}\right) + k$$

References